

03 — Functional Documentation

Complete documentation suite (Markdown → PDF with Mermaid diagrams)

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Source: repository `docs/` directory

Diagrams rendered with Mermaid 11

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Module: Authentication

Purpose

Authenticate users, issue JWTs, expose current user profile, and support password change.

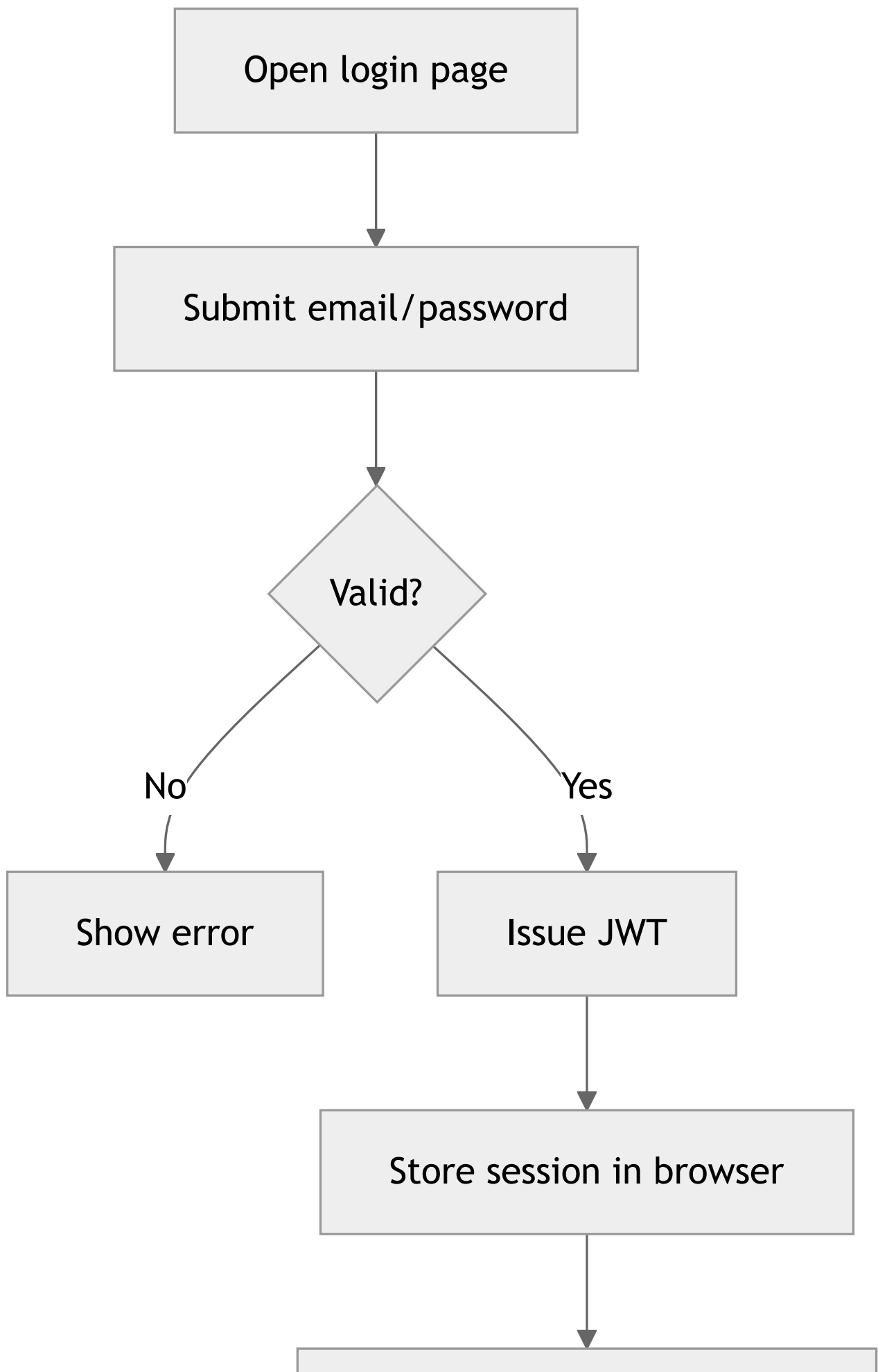
Actors

Any user with an active auth-service account; App Frontend and all portal frontends.

Preconditions

- auth-service running and Mongo reachable
- User exists with password hash; `is_active` true (**Inferred** inactive users rejected — verify service)

Workflow



Navigate dashboard/portal



Optional SSO ?token= to
other app

Business Rules

- Shared `JWT_SECRET` across services
- Optional `MASTER_PASSWORD` can authenticate as any user if set — **must be empty in production**

Validations

Email/password required on login (controller-level).

Permissions

Login public; `/me` and change-password require auth.

Exceptions / Errors

401 invalid credentials; 401/403 on protected routes without/with wrong token.

Related APIs

- `POST /api/auth/login`
- `GET /api/auth/me`
- `POST /api/auth/change-password`

Related DB

User, Role, Portal, Department, CompanyInfo (auth-service)

Related UI

app-frontend `/`, portal `/login` pages

Dependencies

auth-service; MongoDB; frontends

Source

```
auth-service/src/modules/auth/
```

Module: Users & Identity

Purpose

Administer users, departments, roles, portals, and company branding.

Actors

User Manager admins / super admins; auth-service authenticated callers.

Preconditions

Authenticated session with User Manager access gate.

Workflow — assign portal

1. Open User Manager → Portals or User detail
2. Select user
3. Assign `portal_code` + `access_roles`
4. User relaunches App Frontend / portal to pick up JWT claims (**Inferred**: may need re-login)

Business Rules

- Email unique
- DEFAULT_PORTALS seed currently includes **only** `opms`
- Other portals used by apps must be created/assigned manually or via Portals UI seed extensions — **Requires confirmation** of production seed procedure

Permissions

User Manager frontend `hasSuperAdminAccess` ; API routes behind `requireAuth` (finer permission codes **Unknown** vs portal checks)

Related APIs

`/api/users` , `/api/roles` , `/api/portals` , `/api/departments` , `/api/company-info`

Related DB

User, Role, Department, Portal, CompanyInfo

Related UI

user-manager-frontend `/dashboard/*`

Source

```
auth-service/src/modules/users|portals|departments|companyInfo/ user-manager-frontend/
```

Module: Orders & Workflow

Purpose

Create and progress sales orders through departmental statuses to delivery/cancellation.

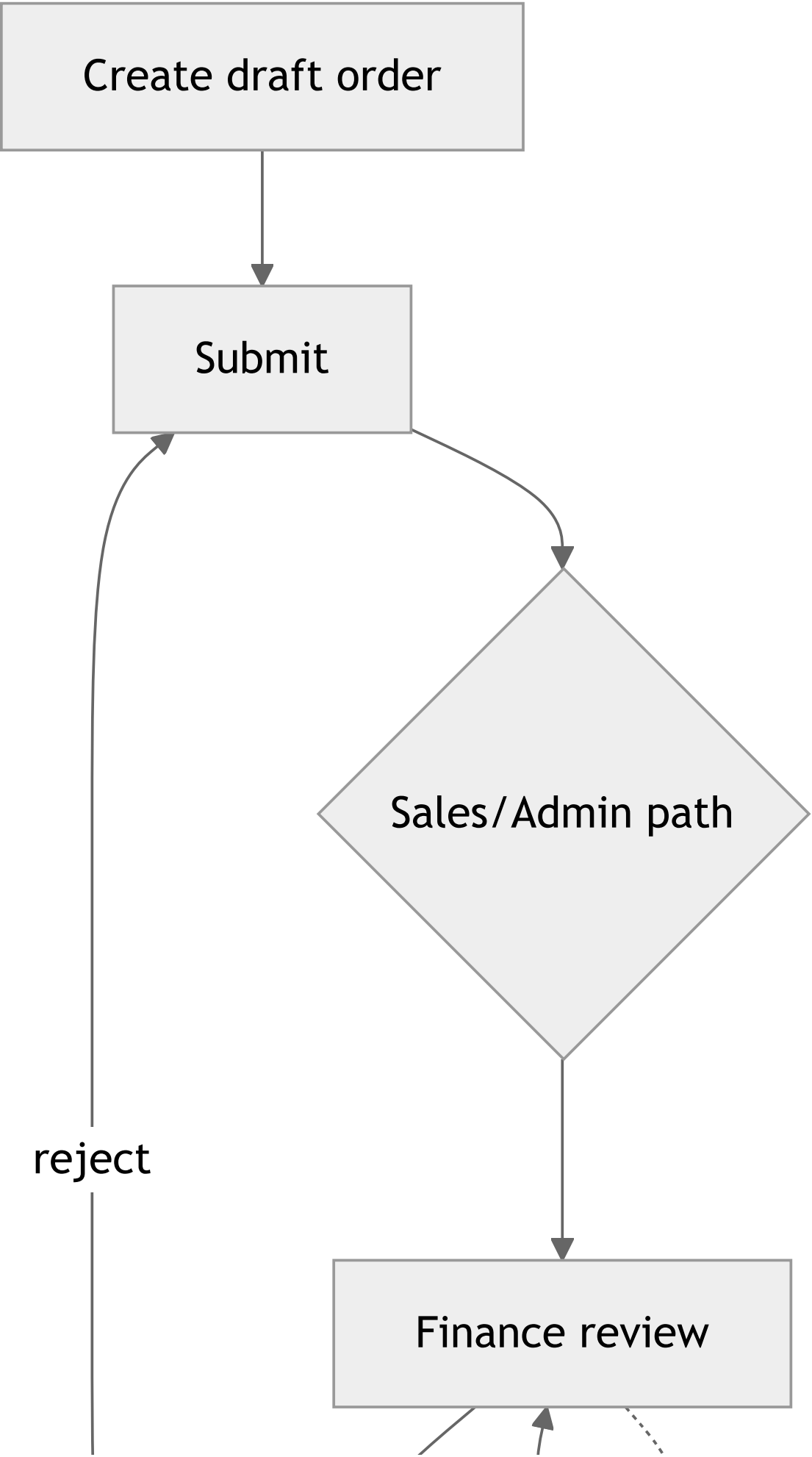
Actors

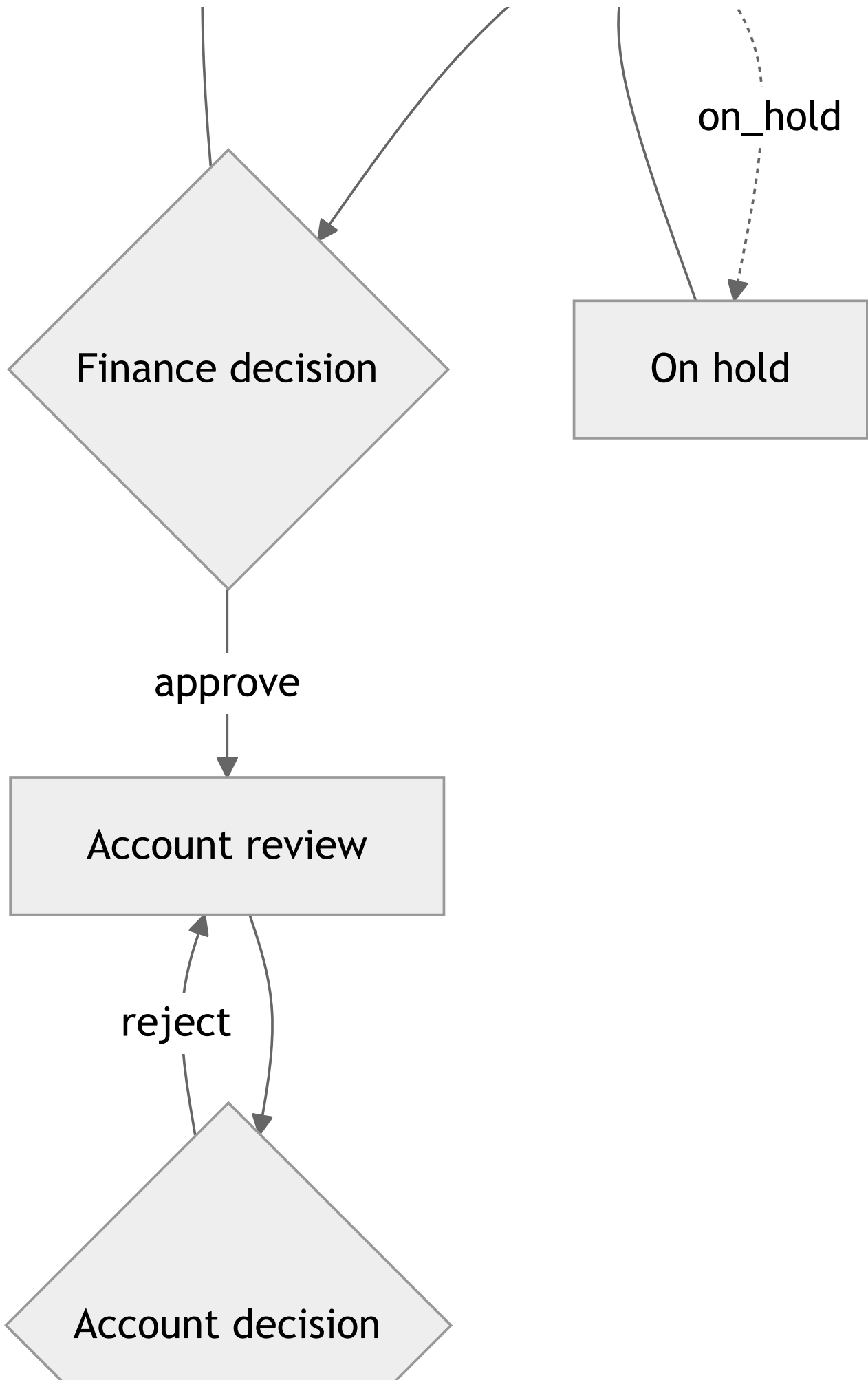
OPMS roles: sales, admin, finance, account, dispatch, super_admin

Preconditions

- OPMS portal access
- Party/product masters available for line items

Workflow





Exact edges: see [Business Rules](#).

Business Rules

- Transition graph enforced
- Flag types can block dispatch/transit/delivery
- Pricing lock after certain statuses
- Soft-delete supported

approve

Validations

Order payload validations in `order-service/controller` (**see service for required fields**).

Permissions

Dispatch

`requireAuth + requireOpmsAccess ; transition permission by department rules.`

State Transitions

See `Order.status` enums and `workflow.transitions.js`.

Related APIs

In transit

`/api/orders/* , /api/orders/:id/transition , /submit , /close-after-full-delivery`

Related DB

`Order`, `OrderWorkflow`, `OrderStatusHistory`, `order_items` embedded

Related UI

`opms-frontend` `/[portal]/orders` , `create-order` , `order/[id]`

delivered

Dependencies

`party/product` data; `notifications/messages` optional

Source

`opms-backend/src/modules/orders/` , `workflow/`

Module: Approvals & Finance / Account

Purpose

Unified order approvals with admin/finance/account signatures; finance queues; due sheets; unbilled tracking; final statements.

Actors

admin, finance, account, super_admin (and sales where allowed)

Workflow (approval document)

1. Create/list OrderApproval for order
2. Admin approve / send-to-finance
3. Finance approve / amend / reject
4. Send-to-account
5. Account approve / amend / reject
6. Resolve-dispatch related actions as exposed by routes

Related APIs

- `/api/order-approvals/*` (approve, reject, send-to-finance, send-to-account, finance-amend, amend, resolve-dispatch, soft-delete)
- `/api/approvals` GET
- `/api/finance/queue` , `/api/finance/summary`
- `/api/order-due-sheets/*`
- `/api/unbilled-orders/*`
- `/api/final-order-statements/*`

Related DB

OrderApproval, OrderAmmendmentUser, OrderDueSheet, UnbilledOrder

Related UI

Order detail approval/finance/account tabs in opms-frontend

Source

```
opms-backend/src/modules/orderApproval|finance|orderDueSheet|unbilledOrder|finalOrderStatement|approvals/implementation_plan.md
```

Module: Dispatch & Transport

Purpose

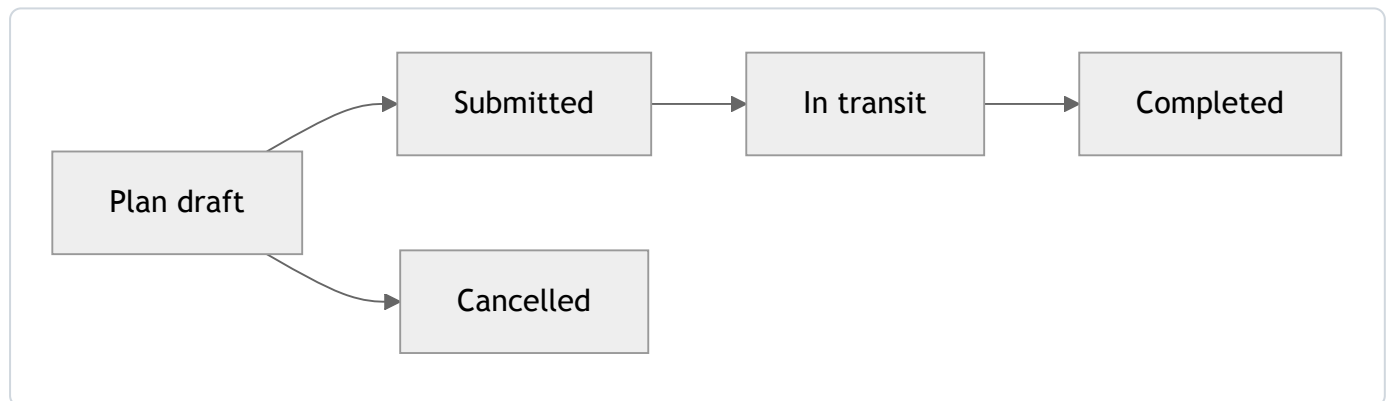
Create dispatches, manage shipments/deliveries/returns, fleet masters, and transport planner.

Actors

dispatch, admin, account, finance, super_admin (per nav/API gates)

Workflow — transport plan

1. List eligible orders
2. Create plan for date/agent
3. Add orders
4. Submit plan
5. Mark packed / generate LR / dispatched / delivered on plan orders
6. Complete or cancel plan



Related APIs

`/api/dispatch`, `/api/transport`, `/api/order-deliveries`, `/api/order-returns`, `/api/vehicles`,
`/api/drivers`, `/api/transport-agents`, `/api/transport-plans`

Related DB

OrderDispatch, TransportShipment, OrderDelivery, OrderReturn, Vehicle, Driver, TransportAgent, TransportPlan, TransportPlanOrder

Note

TransportShipment references `Transporter` model not found in registry — **Unknown** / possible schema debt.

Related UI

Dispatch portal; transport planner pages; fleet routes

Source

```
opms-backend/src/modules/dispatch|transport|orderDelivery|orderReturn|fleet|transportPlanner/
```

Module: Parties & Products

Purpose

Maintain commercial masters: parties, zones, products, kits, brands, manufacturers, party-product rates.

Actors

OPMS admin/finance/account/super_admin primarily; APIs require auth (+ department middleware currently effectively auth-only in party/product).

Workflows

- CRUD + soft-delete/restore/bulk operations
- Zone attach parties and sales persons
- Party product mapping + rate approve
- Google Sheet webhook ingest

Related APIs

party-service: /api/parties , /api/zones , /api/party-products , /api/party-order-products-rate

product-service: /api/products , /api/product-groups , /api/product-subgroups , /api/product-brands ,
/api/product-manufacturers , /api/product-kit-items

Also reachable via opms-backend proxy.

Related DB

Party, Zone, Product, ProductKitItem, ProductGroup, ProductSubgroup, ProductBrand, ProductManufacturer, Batch, PartyProductMapping, PartyProductRate, PartyProductLastRate

Related UI

opms-frontend Party Master / Product Master sections

Source

party-service/ , product-service/

Module: Leads & Quotations

Purpose

CRM pipeline from lead creation through follow-ups, quotations, and conversion.

Actors

lead_manager admin|manager|executive ; limited cross-access from work_planner for lead list.

Lead statuses (enum)

new, assigned, contacted, qualified, unqualified, follow_up, quotation, negotiation, won, lost, converted

Workflow

1. Create lead
2. Assign / change status / qualify
3. Schedule follow-ups
4. Create quotation (admin/manager)
5. Submit/approve/reject quotation
6. Mark won/lost or convert to party/order linkage fields

Jobs

node-cron follow-up reminder digests (FOLLOWUP_REMINDER_*)

Related APIs

/api/leads, /api/lead-masters, /api/quotations, /api/terms-and-conditions, /api/attachments

Related DB

Lead, LeadFollowUp, LeadSource, LeadLostReason, LeadQuotation, FollowUpDigestLog, Terms*

Related UI

lead-manager-frontend /dashboard/leads|quotations|follow-ups|reports

Source

lead-manager-backend/, lead-manager-frontend/

Module: Work Planner

Purpose

Daily field plans (visits/works), approvals, expenses, and per-user settings.

Actors

work_planner admin/manager/executive (+ global admin bypass)

WorkPlan statuses

planned , draft , submitted , approved , rejected , completed

Editable (constants): planned , draft , rejected

Workflow

1. Create plan for date (unique per sales_user+date when active)
2. Add visits/works
3. Submit → manager/admin approve/reject
4. Execute visits (check-in/out/complete)
5. Submit expenses → approve/reject
6. Complete plan / day-end flows in UI

Related APIs

/api/work-planner/* (plans, visits, works, expenses, user-settings, uploads, stats)

Related DB

WorkPlan, WorkPlanVisit, WorkPlanWork, WorkPlanExpense, UserWorkPlannerSettings

Related UI

work-planner-frontend dashboard routes

Source

work-planner-backend/ , work-planner-frontend/

Module: Notifications & Messages

Purpose

In-app notifications (incl. SSE), web push, email, WhatsApp, and communication helpers.

Actors

Authenticated users; internal services calling notification internal endpoints.

Notification workflow

1. Domain service calls internal create / order-transition
2. Persist Notification; enqueue push optionally
3. Client lists or SSE streams; mark read

Message workflow

1. API send / communication trigger
2. Enqueue BullMQ job
3. Worker delivers via Graph/SMTP/WhatsApp
4. Update Message status

Permissions

User routes require auth; **internal** notification POSTs are unauthenticated in code — rely on network trust.

Related APIs

notification-service /api/notifications , /api/push/* , /api/subscribe

message-service /api/messages , /api/emails , /api/auto-emails , /api/communication

Related DB

Notification, PushSubscription, Message

Related UI

NotificationBell across frontends; communication tabs in OPMS order UI

Source

notification-service/ , message-service/

Module: Attachments & Files

Purpose

Upload files to external file API; store metadata; view/download via redirects.

Actors

Authenticated OPMS / lead / work-planner users (note weaker auth on some routers — see security)

Workflow

1. Multipart upload to attachments endpoint
2. File Management API stores binary
3. Attachment document saved with entity_type/entity_id
4. View/download via `/api/files/:fileId/view|download`

Soft-delete

Supported with restore on OPMS attachments (permission middleware).

Related APIs

`/api/attachments` , `/api/files/:fileId/view|download`
work-planner public-ish attachment view path

Related DB

Attachment

Dependencies

`FILE_MANAGEMENT_API_URL` , `FILE_MANAGEMENT_API_KEY`

Source

`opms-backend/src/modules/attachments|files/` , lead/work-planner attachment modules

Module: Dashboards & Activity

Purpose

Role-scoped KPI summaries and activity feed.

Related APIs

- `GET /api/dashboard/admin|sales|finance|dispatch|account|super`
- `GET /api/activity`
- Order stats: `GET /api/orders/stats` , workflow-context
- Transport planner stats
- Work planner / lead manager dashboard stats on their services

Related UI

Portal overview pages; App/User Manager KPI cards; Lead/Work dashboards

Source

`opms-backend/src/modules/dashboard|activity/`